Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (ODD)**	**Course Name**	**PE-I - Geo-Intelligence for Smart IoT Devices Lab**
Semester	**5**	**Course Code**	**23IOT1523**
Roll No	**64**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**01**
Course Outcome	Apply and demonstrate the use of proprietary and open-source GIS tools (e.g., QGIS) for creating, visualizing, and managing spatial datasets.
Aim	Install/verify QGIS, explore interface, and understand open-source vs proprietary GIS features.
Problem Definition	The task is to install and verify QGIS on a computer system, explore its user interface to become familiar with the essential tools, and understand the differences between open-source GIS software and proprietary GIS platforms. (Discuss about 10 open source and proprietary software.)
Theory (100 words)	A Geographic Information System (GIS) is a computer-based system that integrates hardware, software, and data for capturing, storing, analyzing, and displaying spatial or geographic information. It is widely used in urban planning, disaster management, transportation, agriculture, natural resource monitoring, and environmental protection. GIS software is generally categorized into open-source and proprietary types. Open-source GIS software, such as QGIS, is free, flexible, and allows users to modify the source code for customization. It is cost-effective and popular among students, researchers, and small organizations. However, it usually depends on community support and requires technical knowledge for troubleshooting or adding new functions. Proprietary GIS software, such as ArcGIS, is commercial and requires paid licenses or subscriptions. It provides advanced enterprise-level tools, professional vendor support, training programs, and frequent updates, making it reliable for government bodies and large industries. By studying both categories, it becomes clear that open-source software offers affordability and flexibility, while proprietary solutions focus on stability, advanced features, and official support, helping users choose the right platform based on purpose and budget.

Aspect	Open-Source GIS (e.g., QGIS)	Proprietary GIS (e.g., ArcGIS)
Cost	Free to use, no license fee.	Paid license or subscription required.
Source Code	Open and modifiable by users.	Closed-source, cannot be modified.
Learning Curve	May require technical knowledge.	Easier with vendor training and manuals
Features	Good for general GIS needs, flexible for research	Advanced, enterprise-level features for industries.
Updates	Frequent updates from global community.	Regular professional updates and bug fixes from vendor.

- **Open-Source GIS Software**
  1. **QGIS –** A popular open-source desktop GIS software known for its user-friendly interface, wide range of plugins, and strong community support. It is suitable for mapping, spatial analysis, and data visualization.
  2. **GRASS GIS –** A powerful open-source GIS with advanced geospatial modeling, raster analysis, and scientific research capabilities. It is widely used in academia and environmental studies.
  3. **uDig –** A lightweight, Java-based desktop GIS application designed for ease of use. It focuses on data viewing, editing, and integration with web services.
  4. **gvSIG –** An open-source GIS platform offering tools for desktop and mobile GIS applications. It is widely used for cartography, spatial data analysis, and project management.
  5. **PostGIS –** A spatial database extender for PostgreSQL that adds support for geographic objects. It enables advanced spatial queries and geospatial data management.

	6. **GeoServer –** An open-source server for sharing geospatial data using web standards like WMS, WFS, and WCS. It allows easy publishing and integration of GIS data over the web. 7. **Leaflet –** A lightweight JavaScript library for building interactive web maps. It is simple, mobile-friendly, and widely used in web development. 8. **MapServer –** An open-source platform for publishing spatial data and interactive web maps. It supports OGC standards and is suitable for web GIS applications. 9. **SAGA GIS –** A GIS software specialized in geoscientific analysis, terrain modeling, and raster data processing. It is highly efficient for spatial data research. 10. **Whitebox GAT –** An open-source GIS focused on advanced geospatial analysis and remote sensing. It is often used for environmental and hydrological modeling. ➢ **Proprietary GIS Software** 1. **Esri ArcGIS –** A leading proprietary GIS platform offering advanced mapping, spatial analysis, and enterprise-level solutions. Widely used in government, business, and research for its reliability and vendor support. 2. **Autodesk AutoCAD Map 3D –** A GIS and mapping extension of AutoCAD integrating CAD and GIS data. It is useful for infrastructure planning, utilities, and engineering projects. 3. **MapInfo Pro –** A desktop GIS software focused on spatial data visualization and business analysis. Commonly used in market research, urban planning, and location intelligence. 4. **Manifold GIS –** A fast, efficient GIS tool that integrates mapping, spatial analysis, and database management. Known for handling large datasets with high performance. 5. **Bentley MicroStation –** A CAD-based software with GIS capabilities for infrastructure design and geospatial modeling. Extensively used in engineering, construction, and transportation sectors.

	6. **Global Mapper –** A cost-effective GIS application supporting a wide range of data formats and terrain analysis. It is popular for remote sensing and LiDAR processing. 7. **CARTO –** A cloud-based GIS and location intelligence platform designed for web mapping and data visualization. It is widely used in business analytics and decision-making. 8. **SuperGIS Desktop –** A proprietary GIS software developed by SuperGeo Technologies, offering tools for mapping, editing, and spatial analysis. It is popular in Asia for government and academic use. 9. **Intergraph G/Technology –** A specialized GIS solution for managing utility and telecommunications networks. It supports asset management, network planning, and operations. 10. **Cadcorp SIS –** A GIS platform for spatial data analysis, mapping, and web GIS services. It is widely used in local governments, emergency services, and public sectors.
Procedure and Execution (100 Words)	Implementation Steps: 1. QGIS Installation ➢ Download: Go to the QGIS website https://qgis.org/ or https://qgis.org/download/ and download either the latest Long-Term Release (LTR) version or current version of the software that fits your operating system (Windows, macOS, Linux). ➢ Installation: Run the installer, follow the default setup, and ensure dependencies (like GDAL, GRASS, and Python bindings) are installed. ➢ Verification: • Start QGIS → Open a sample dataset (e.g., shapefile, GeoTIFF). • Confirm that the map layers load properly. • Check that tools are working (e.g., zoom, pan, attribute table, coordinate system) too. • Confirm plug-in manager is working.

	Exploring the QGIS User Interface – Key Components 1. **Menu Bar** o Provides structured access to project settings, data handling, layer operations, vector/raster tools, and plugins. o Works as the main control panel for all core functions of QGIS. 2. **Toolbars** o Contain shortcut buttons for frequently used actions like navigation, zooming, feature selection, measuring, and editing. o Users can rearrange, hide, or customize toolbars as per workflow. 3. **Browser Panel** o Helps in exploring spatial data from local files, databases, and remote servers. o Supports drag-and-drop loading of datasets directly into the project. 4. **Map Display Area (Map Canvas)** o The central workspace where loaded spatial data is visualized and edited. o Supports real-time interaction with features and map navigation. 5. **Layers Panel** o Displays all active map layers in the project and controls their visibility and order. o Allows changing symbology, transparency, grouping, and layer properties.
	Stepwise Screenshots with steps:

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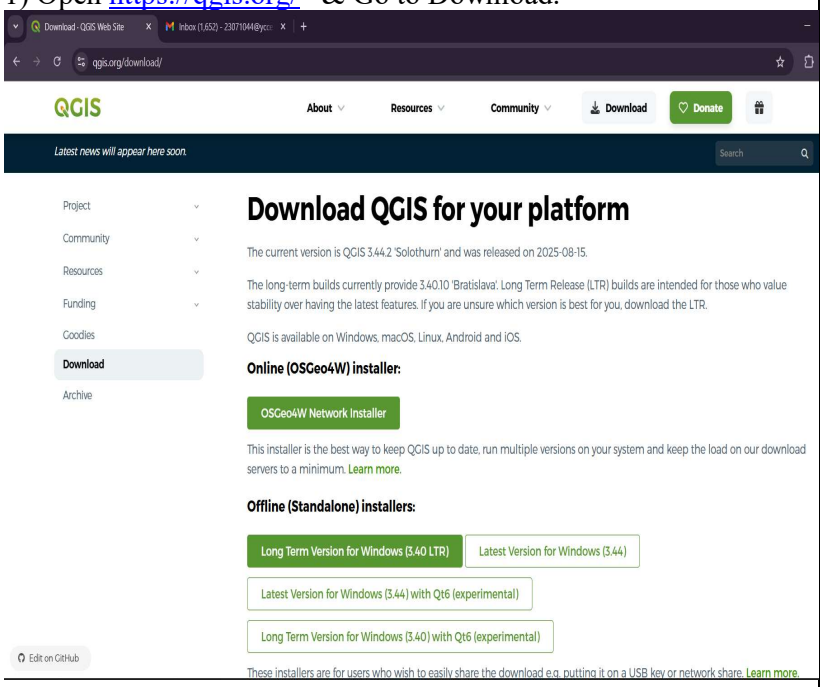
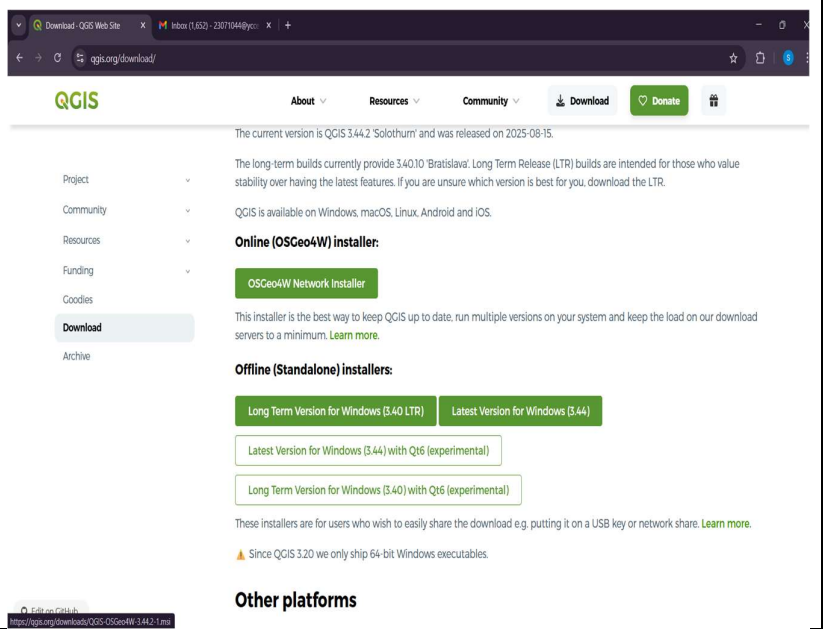
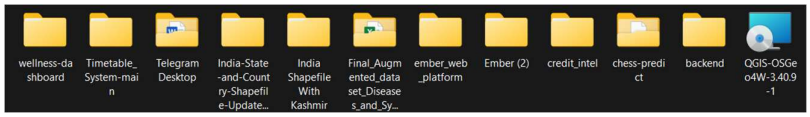
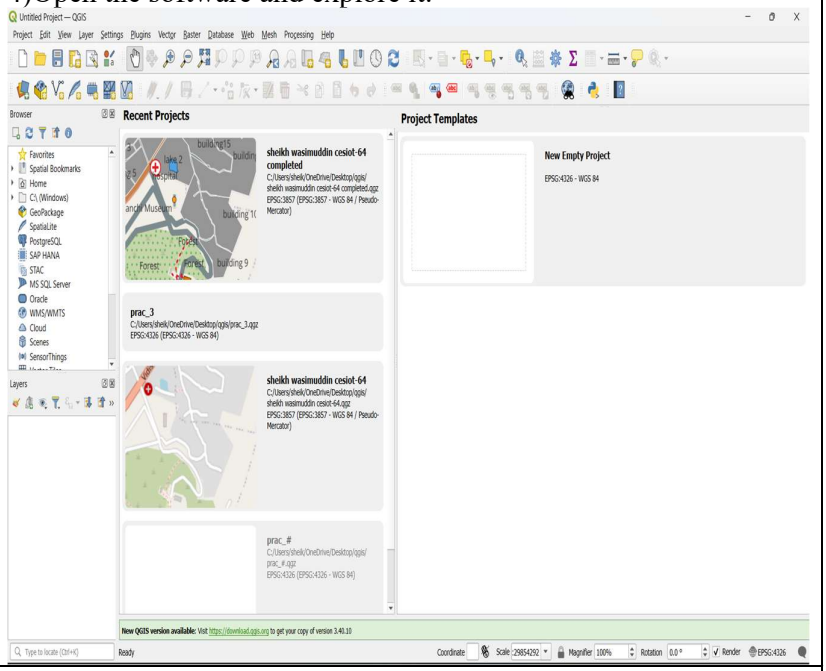
1) Open https://qgis.org/ & Go to Download.



2)Click on Offline InstallerVersion which is suitable for your os.



	3)Install the Installer Package.  4)Open the software and explore it. 
Output Analysis	During this exploration, I studied different GIS platforms including QGIS, GRASS GIS, uDig, gvSIG, PostGIS, GeoServer, Leaflet, MapServer, SAGA GIS, and Whitebox GAT. These tools are open-source, freely available, and supported by global developer communities, which makes them highly flexible and customizable. Each has its own strengths—for example, QGIS is known for its plugins and ease of use, GRASS GIS for advanced raster analysis, PostGIS for database integration, and Leaflet for web mapping. Although they sometimes require more technical knowledge compared to commercial software, they provide powerful functionality without licensing costs. From my perspective, QGIS stands out as the most balanced option because it combines a user-friendly interface, cross-platform support, plugin integration, and strong community resources, making it highly suitable for both beginners and professionals.
Conclusion	The experiment was successfully completed by installing and exploring QGIS, which helped in understanding its interface,

	tools, and functionalities. Along with this, a detailed study was carried out on the differences between open-source and proprietary GIS software, highlighting their advantages, limitations, and key features. This exploration provided clarity on how open-source platforms like QGIS offer cost-free flexibility and customization, while proprietary software focuses on advanced enterprise solutions and vendor support. Overall, the activity enhanced practical knowledge of GIS tools and built a strong foundation for selecting the right platform based on requirements.
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Date	**22/07/2025**